

Graphs of Sine and Cosine Functions

ANSWER KEY



Amplitude, period, and midline

- 1 For the function $y = 3\sin(2x)$, state:
a the amplitude $= 3$ **b** the period $= \pi$ **c** the range $[-3, 3]$
 - 2 For the function $y = -2\cos\left(\frac{x}{2}\right) + 1$, state:
a the amplitude $= 2$ **b** the period $= 4\pi$
c the midline $y = 1$ **d** the range $[-1, 3]$
 - 3 For $y = \sin\left(x - \frac{\pi}{3}\right)$, state the phase shift and the period.
Phase shift: $\frac{\pi}{3}$ to the right. Period: 2π .
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Reading and writing equations from graphs

- 4 The graph below shows one sinusoidal function. State its amplitude and period, then write an equation for the function.
Amplitude 2, period 2π . Equation: $y = 2\sin x$.
- 5 Write an equation of a cosine function with amplitude 4, period π , reflected across the x -axis, with no phase or vertical shift.
 $y = -4\cos(2x)$ (period $\frac{2\pi}{b} = \pi$ gives $b = 2$).
- 6 The depth of water at a harbour entrance is modeled by $d(t) = 2.5\sin\left(\frac{\pi}{6}t\right) + 7$ metres, where t is in hours after midnight.
a State the maximum and minimum depths. Maximum $7 + 2.5 = 9.5$ m; minimum $7 - 2.5 = 4.5$ m.
b State the period of the tide cycle. Period $= \frac{2\pi}{\pi/6} = 12$ hours.
c Find the depth at $t = 3$. $d(3) = 2.5\sin\left(\frac{\pi}{2}\right) + 7 = 9.5$ m.